Pranav Reddy Danda

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Education

Webster University, M.S. in Cybersecurity with Emphasis in Artificial Intelligence – St. Louis, MO, USA — Aug 2024 – May 2026

Matrusri Engineering College, Osmania University, B.E. in Electronics & Communication Engineering – Hyderabad, TG, India — 2019 – 2023

Experience

Research Assistant, Webster University – St. Louis, MO, USA — Jan 2026 – May 2026

- Architected and fine-tuned transformer-based NLP models using PyTorch and Hugging Face Transformers for domain-specific tasks (LLMs, NLP, attention mechanisms)
- Designed and executed distributed training workflows on Linux servers with GPU acceleration and efficient data pipelines
- Deployed and managed cloud experiments on Chameleon Cloud using bare-metal servers and virtual machines.
- Implemented experiment tracking (MLflow, Weights & Biases) for reproducible model evaluation and tuning
- Integrated cloud services (AWS, GCP) and containerization (Docker) into ML deployment pipelines
- Optimized model performance with hyperparameter tuning, cross-validation, and algorithmic benchmarking

Network Security Intern, Bharat Sanchar Nigam Limited (BSNL) – Hyderabad, India — May 2022 – May 2023

- Performed vulnerability assessment, penetration testing support, and network security hardening
- Diagnosed and resolved routing issues using TCP/IP and network protocols
- Built automated Python tooling for logging, data synchronization, and secure key encryption
- Applied threat analysis and incident response practices in enterprise-grade infrastructure

Academic Projects

AI-Driven Intruder Detection (IoT & Computer Vision)

- Built an intrusion detection system using motion sensors, discord API, and computer vision techniques using open-cv.
- Implemented real-time alerts with model inference for face detection and edge device integration with pytorch.
- Demonstrated applied deep learning and anomaly detection capabilities.

Hybrid Hexapod Robot with AI-Based Disease Classification

- Designed a robotic platform for autonomous agricultural monitoring using Node.js and PyQt library.
- Designed the robot in fusion 360 and 3d printed it
- Implemented image classification via RNNs and transfer learning on Raspberry Pi architecture
- Integrated feature extraction, model evaluation, and predictive analytics
- Project can be found here https://drive.google.com/file/d/1LG9y35iKhGacuAVbIrhXHc2bpjcnMLFp/view?usp=sharing

Publications

Danda, P. R., & Li, X. (2026). Energy Profiling of Pruned LLMs During Python Code Generation. IGSC 2026, Canandaigua, NY.

Certifications

Google Cybersecurity Professional Certificate – Coursera

Google Advanced Data Analytics – Coursera

CompTIA Security+ (In Progress)

Skills

AI Deployment & Tools

- Deep Learning & Neural Networks,
- Transformer Models (LLMs),Natural Language Processing (NLP)
- PyTorch, TensorFlow, Keras ,HuggingFace, Transformers library,OpenCV
- Docker, Kubernetes
- Cloud Platforms(AWS / GCP / Azure/ Chameleon Cloud)
- ROS, RViz, Gazebo, and PCL.
- Fusion 360,SolidWorks,AutoCAD

Cybersecurity & IT Security

- Threat Analysis & Vulnerability Assessment
- Network Security & Intrusion Detection
- Penetration Testing
- Zero Trust Architecture
- Encryption & Identity Access Management

Programming & Systems

- Python, SQL, C++, JavaScript
- Linux Administration
- Git / GitHub / Version Control
- Data Structures & Algorithms
- REST APIs & Backend Logic